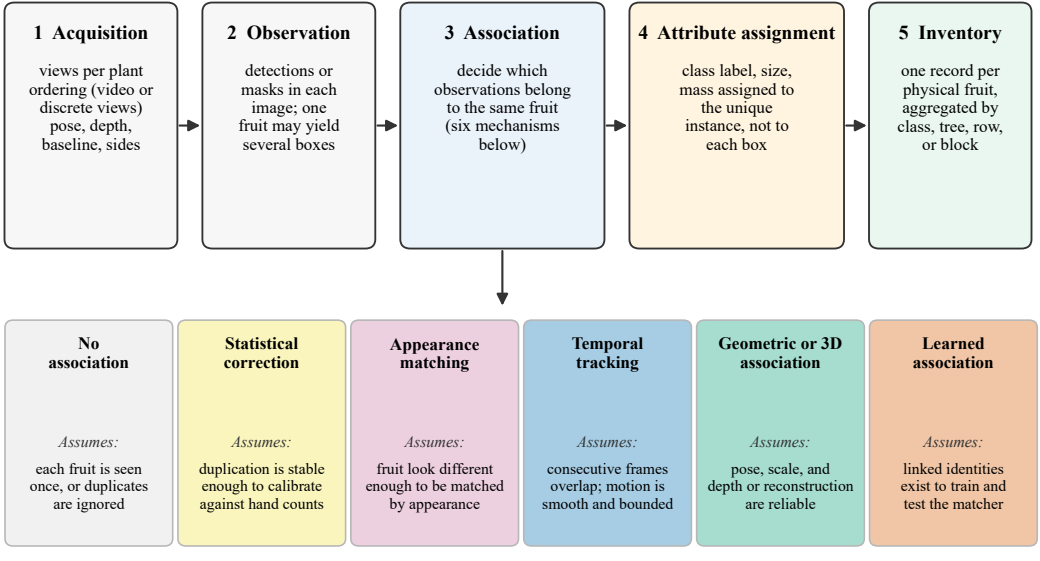

1 Acquisition
views per plant ordering (video or discrete views) pose, depth, baseline, sides
2 Observation
detections or masks in each image; one fruit may yield several boxes
3 Association
decide which observations belong to the same fruit (six mechanisms below)
4 Attribute assignment
class label, size, mass assigned to the unique instance, not to each box
5 Inventory
one record per physical fruit, aggregated by class, tree, row, or block
No association
Assumes:
each fruit is seen once, or duplicates are ignored
Statistical correction
Assumes:
duplication is stable enough to calibrate against hand counts
Appearance matching
Assumes:
fruit look different enough to be matched by appearance
Temporal tracking
Assumes:
consecutive frames overlap; motion is smooth and bounded
Geometric or 3D association
Assumes:
pose, scale, and depth or reconstruction are reliable
Learned association
Assumes:
linked identities exist to train and test the matcher